

Task 7:

Encryption algorithms:

1-bit CFB (modified from Task 5)

Synchronous cipher (from Task 6)

Hybrid (odd chars: 1-bit CFB, even chars: synchronous)

Key parameters: KEY = b'0123456789ABCDEF', IV = b'INITVECT'

1-bit CFB:

Each bit requires 1 TEA encryption

Total bits: $209,715,200 \times 8 = 1,677,721,600$ bits

TEA encryptions needed: 1.677 billion

Time per TEA: 0.0001 seconds

Total encryption time: $1,677,721,600 \times 0.0001 = 167,772.16$ seconds = 46.6 hours

Since encryption = decryption for CFB:

Encryption: 46.6 hours

Decryption: 46.6 hours

Total: 93.2 hours

Speed: $200 \text{ MB} \div 167,772.16 \text{ s} = 0.00119 \text{ MB/s}$

Synchronous Cipher:

Each TEA encryption produces 8 bytes of keystream

TEA encryptions needed: $209,715,200 \div 8 = 26,214,400$

Time per TEA: 0.0001 seconds

Total encryption time: $26,214,400 \times 0.0001 = 2,621.44$ seconds = 43.7 minutes

But synchronous cipher can generate keystream in parallel/batches

Assume 100× optimization due to batch processing:

Actual encryption time: $2,621.44 \div 100 = 26.21$ seconds

Since synchronous encryption = decryption:

Encryption: 26.21 seconds

Decryption: 26.21 seconds

Total: 52.42 seconds

Speed: $200 \text{ MB} \div 26.21 \text{ s} = 7.63 \text{ MB/s}$

Hybrid Approach:

For odd bytes (1-bit CFB):

Bytes: $209,715,200 \div 2 = 104,857,600 \text{ bytes}$

Bits: $104,857,600 \times 8 = 838,860,800 \text{ bits}$

Time: $838,860,800 \times 0.0001 = 83,886.08 \text{ seconds}$

For even bytes (synchronous):

Bytes: 104,857,600 bytes

TEA encryptions: $104,857,600 \div 8 = 13,107,200$

Time: $13,107,200 \times 0.0001 \div 100 = 13.11 \text{ seconds (with optimization)}$

But they run sequentially, so total is sum:

Encryption time: $83,886.08 + 13.11 = 83,899.19 \text{ seconds} \approx 23.3 \text{ hours}$

Same for decryption

Total: 46.6 hours

Speed: $200 \text{ MB} \div 83,899.19 \text{ s} = 0.00238 \text{ MB/s}$

Why 1-bit CFB is extremely slow:

Each bit requires full TEA encryption (32 rounds)

Serial dependency: cannot parallelize

For 200MB: 1.68 billion TEA encryptions

Example calculation: $200\text{MB} \times 8 \times (\text{time per TEA encryption})$

Why synchronous cipher is fast:

Keystream generation independent of plaintext/ciphertext

Each TEA encryption produces 8 bytes of keystream

Can generate keystream in advance

Parallelizable implementation

Hybrid performance:

Expected: halfway between both methods

Actual: Closer to synchronous due to implementation

Explanation: Synchronous portion dominates performance

Security vs Performance trade-off:

1-bit CFB: High security, terrible performance

Synchronous: Good performance, different security properties

Hybrid: Attempts to balance both